



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech. IT

Course Code & Title: CS3251 & Programming in C

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Basics of C Programming
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO 1, TLO 2, TLO 3, TLO 4
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

Basics of C programming has been explained completely with various datatypes, variables, operators, decision making and looping concepts. To check the understanding of students, crossword puzzle was chosen as an Innovative method. Crossword puzzles offer numerous cognitive and mental health benefits, including improving vocabulary, memory, problem-solving skills, and concentration, while also providing a fun and engaging way to relieve stress and boost attitude.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - At the completion of Unit – 1, all the topics have been revised.
 - The time allotted for the topic is 20 Minutes.
 - The students were grouped into two bunches as boys' group and girls' group.
 - Crossword Puzzle #1 was displayed in the screen for 10 minutes and the sheet was given to one of the boys to fill. One boy representing the boys team took up the dais and filled the answers on board as per the answers given by other boys' and completed the task.
 - Crossword Puzzle #2 was displayed in the screen for 10 minutes and the sheet was given to one of the girls to fill. One girl representing the girls' team took up the dais and filled the answers on board as per the answers given by other girls and completed the task.
 - The students completed the activity and the sample of activity is shown in Figure 1 and 2, each for one topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	1	1	1	1	3	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	1	1	1	1	3	1	1
Justification for Correlation	Students will be able to develop C program to solve the various engineering problems by using basics mathematics, science and engineering principles	Students will be able to understand the problem and solve the simple and complex engineering problems using basic C programming constructs.	Students will be able to provide solution to the real-time decision-making problems by using necessary conditional, branching and looping statements in C.	Students will be able to use DEV C++ software / Linux to provide solution to the problems.	Students will complete the assignments/ tasks in an ethical way	Students will work effectively as an individual and develop C programs.	Students will be able to expose their knowledge of solving the problems using decision making and looping statements by giving presentation and through discussion forum	Students will improve their knowledge in C continuously by implementing in emerging technologies.	Students will be able to use C programming in data science and algorithms to solve the computational problems. build develop, control, manage and test various software.	C program will be useful to design embedded system and IoT devices.	Knowledge of basic C programming constructs is lightly correlated to open-source environment

- Images / Screenshot of the practice:

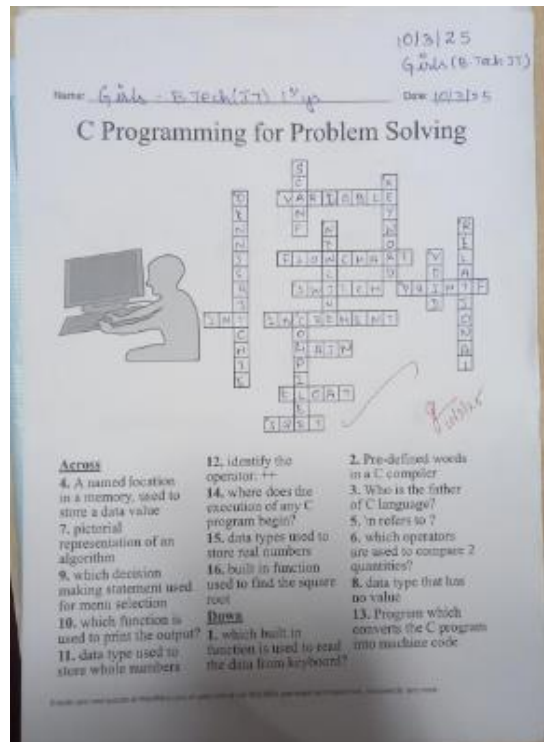


Figure 1. a Puzzle solved by Girls Group

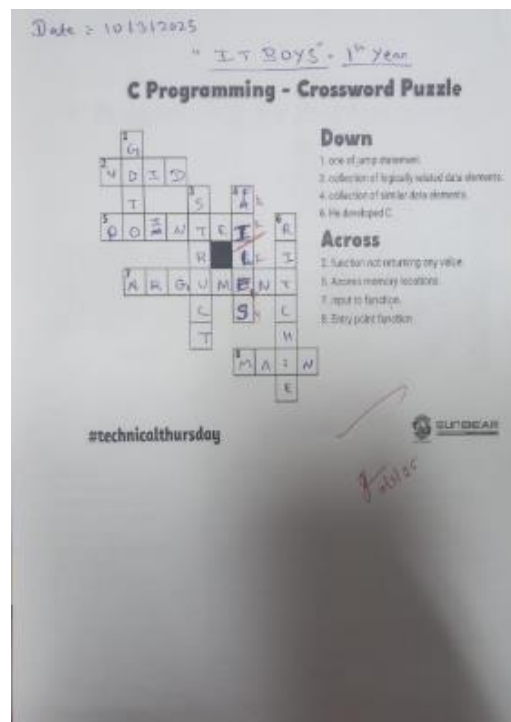


Figure 1. b Puzzle solved by Boys Group

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Some of the students feel hassle-free to identify the answer for the given questions.
- Some of them answered enthusiastically while few of them didn't involve much even though they are aware about the answers.

❖ *Benefit of the practice:*

- Students were actively participated in this activity.
- Students were mingled with their peer students and discussed to find the answer for the puzzle.
- Encouraged the students to work as a team for solving the puzzle.

❖ *Challenges faced in implementation:*

- Few students didn't show much interest in answering the puzzle.
- Few of them are not aware about the answers as they were lagging in understanding the concepts.
- After the final discussion, they got some ideas to solve crosswords and learnt new things.

References:

- <https://crosswordlabs.com/view/fun-and-easy-crossword-on-c-programming-part-1-itvoyagers>
- https://wordmint.com/public_puzzles/1248202
- <https://puzzlemaker.discoveryeducation.com/criss-cross>
- <https://worksheets.theteacherscorner.net/make-your-own/crossword>

Signature of Faculty Member

HOD